# COS 101
# Introduction to Computing

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# Outline

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# What is a Computer?

**Definition:** A computer is an electronic device that:

- Takes **data** as **input**
- **Processes** the **data** under the control of a *program* (set of instructions)
- Produces an **output**
- The entire system is called a **computer system**

# History of Computers: A Brief Timeline

**Computing Evolution:**

- Born out of a need for efficient number-crunching
- Punch-card based computers started around the late 19th century
- Modern smartphones possess more computing power than early computers

## Early Milestones in Computing (1801 - 1937)

- **1801:** *Joseph Marie Jacquard's loom* uses punch cards for weaving fabric designs
- **1822:** *Charles Babbage* conceives the first steam-driven *calculating machine* (though incomplete)
- **1890:** *Herman Hollerith* develops a *punch card system* to calculate the U.S. Census
- **1936:** *Alan Turing* presents the concept of a *universal computing machine* (Turing Machine)

## Mid-20th Century Developments (1939 - 1958)

- **1941:** *Atanasoff* and *Clifford Berry* create a computer with memory for *simultaneous equation solving*
- **1943-1944:** *ENIAC*, the first large-scale digital computer, is built by Pennsylvania professors, *John Mauchly* and *J. Presper Eckert*
- **1946:** *UNIVAC* by *John Mauchly* and *J. Presper Eckert* becomes the first commercial computer for government/business use
- **1953-1958:** *Grace Hopper* develops *COBOL*, and the *integrated circuit* (computer chip) is invented

## Personal Computers & Major Inventions (1964 - 1981)

- **1964:** *Douglas Engelbart* invents the *computer mouse* and *GUI*
- **1969:** *UNIX* operating system developed
- **1971:** *Alan Shugart* invents the *floppy disk*
- **1975:** *Paul Allen* and *Bill Gates* write software for the Altair, using BASIC. Altair 8080 sparks the *personal computer* revolution and *Microsoft* is born
- **1976:** *Steve Jobs* and *Steve Wozniak* start Apple Computers on April Fool's Day and roll out the  Apple I, the first computer with a *single-circuit board*, according to Stanford University
- **1981:** *IBM* introduces its first personal computer using *MS-DOS*

## The Internet Era & Modern Computing (1990 - Present)

- **1990:** *Tim Berners-Lee* invents *HTML* and the World Wide Web is born
- **1996:** *Sergey Brin* and *Larry Page* develop the Google search engine
- **2016:** The first reprogrammable quantum computer was created, enabling the programming of new algorithms into a single system for the first time
- **2017-Present:** DARPA develops Molecular Informatics using molecules as computers

# Generations of Computers

In summary, there are five Generations of Computers:

- First Generation (1940–1956): Vacuum tubes
- Second Generation (1956–1963): Transistors
- Third Generation (1964–1971): Integrated circuits
- Fourth Generation (1971–present): Microprocessors
- Fifth Generation (present & future): Artificial intelligence

# Characteristics of the computer system

- **Speed:** Performs millions of instructions per second, operating in microseconds or nanoseconds
- **Accuracy:** Executes calculations with 100% precision, errors occur only with inconsistent or inaccurate data
- **Diligence:** Maintains consistent performance without fatigue, capable of millions of operations with accuracy
- **Versatility:** Can handle a variety of tasks with equal accuracy and efficiency
- **Reliability:** Produces consistent results for the same input every time
- **Automation:** Operates tasks automatically without the need for manual intervention
- **Memory:** Stores data in primary (built-in) and secondary (removable) storage devices like CDs and pen drives

# Data and Information

- **Data:**

  Data  is a *collection of unorganized facts & figures* and does not provide any further information regarding patterns, context, etc. Hence data means "unstructured facts and figures".

- **Information:**

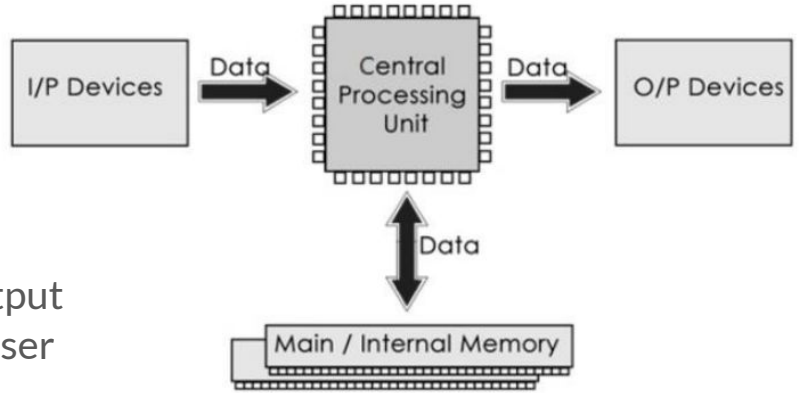
  Information is *structured data* i.e. *organized, meaningful and processed data*. To process the data and convert into information, a computer is used.

# Functions of Computers

- **Receiving Input:**
  - Data is fed into computer through various input devices like keyboard, mouse, digital pens, etc.
  - Input can also be fed through devices like CD-ROM, pen drive, scanner, etc.
- **Processing the information**
  - Operations on the input data are carried out based on the instructions provided in the programs.
- **Storing the information**
  - After processing, the information gets stored in the primary or secondary storage area.
- **Producing output**
  - The processed information and other details are communicated through output devices like monitor, printer, etc

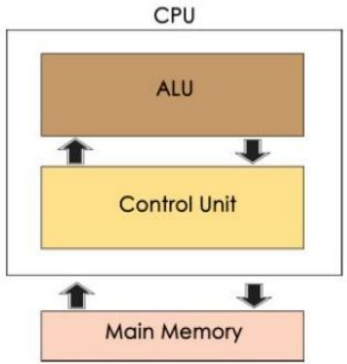
# Components of a Computer System

- **Key Components:**
  - Central Processing Unit (CPU)
  - Input Devices
  - Output Devices
- **Process Overview:**
  - Input devices provide data to the CPU
  - CPU processes data and generates output
  - Output devices display results to the user
  - Data is stored in computer memory



# Central Processing Unit (CPU)

- Definition: The "brain" of the computer that controls operations of all parts
- Components:
  - **Arithmetic Logic Unit (ALU):**
    - Processes data, performs calculations, and comparisons
    - Non-numeric processing and decision-making occur here
  - **Control Unit:**
    - Extracts instructions and directs CPU operations
    - Supervises data flow within the CPU



## Functions of the Control Unit

- Controls all activities of the computer.
- Supervises and directs the flow of data within the CPU.
- Transfers data to the ALU and results to memory.
- Fetches results from memory to output devices.

## Memory Unit

- Definition: Stores data, instructions, and results
- Types: **Primary** and **Secondary**
- Unit of Memory: Byte (**1 Byte = 8 Bits**).

# Input Devices

**Definition:** Devices that provide data input to the computer.

**Examples:**

- Keyboard
- Mouse
- Trackballs
- Digital pens
- Scanners
- Barcode readers
- Voice recognition systems
- Touch screens

# Output Devices

**Definition:** Devices that that display output to the user

**Examples:**

- Monitors
- Printers
- Speakers (sound systems)

# Primary Memory

- Definition: Main memory present on the motherboard
- Types:
  - **Random Access Memory (RAM):**
    - Volatile, temporary storage
    - Data is lost when powered off
    - Types: Static RAM (SRAM), Dynamic RAM (DRAM)
  - **Read Only Memory (ROM):**
    - Non-volatile, permanent storage
    - Data remains when powered off
    - Types: PROM, EPROM, EEPROM, Flash memory





# RAM vs. ROM

| Feature | RAM | ROM |
|---|---|---|
| **Volatility** | Volatile (temporary) | Non-volatile (permanent) |
| **Data Loss** | Lost when power is off | Retained when powered off |
| **Storage Capacity** | Small capacity | High capacity |
| **Processing Speed** | High speed | Low speed |
| **User-defined** | Yes (modifiable) | No (read-only) |
| **Cost** | High | Cost-effective |
| **Types** | SRAM, DRAM | PROM, EPROM, EEPROM, Flash |

# Secondary Memory

**Definition:** Supplementary storage for larger data sets

**Characteristics:** Non-volatile, permanent storage

**Examples:**

- Compact disks
- Floppy disks
- Pen drives
- External hard drives

# Types of Computers (Based on Operation)

- **Analog Computers:** Measure physical quantities; now nearly extinct
- **Digital Computers:** Use binary (0 and 1), programmable, suitable for complex computations
- **Hybrid Computers:** Combine analog and digital functionalities for process control

# Types of Computers (Based on Size and Processing Power)

- **Microcomputers:** Uses microprocessor as CPU; also called personal computers (PCs)
  - Smaller in size
  - Input: Keyboard, mouse
  - Output: Monitor
  - Examples: Desktops, laptops, notebooks, tablets, Personal Digital Assistants (PDAs)
- **Minicomputers:** Multi-user computers supporting 10 to 100 users simultaneously.
  - Positioning: Intermediate in size and processing capacity between microcomputers and mainframes.
  - Use: Mid-range systems or workstations for businesses and organizations

- **Super Mini Computers:** 32-bit word length, Memory: 1 M byte to 8 M bytes, Speed: Nearly 2 million instructions per second
  - Applications: Research, development, process control in commercial organizations.
- **Mainframe Computers:** Powerful multi-user computers, capable of supporting 100 to 1000+ users simultaneously
  - Applications: Used in large organizations for processing bulk data and critical applications like ERP
- **Supercomputers:** Extremely fast computers capable of hundreds of millions of instructions per second.
  - Applications: Quantum physics, weather forecasting, molecular theory, scientific research

# Concept of Hardware and Software

- **Hardware:** Physical components of a computer system including the:
  - Input Devices: Keyboard, Mouse, Joystick, Scanner
  - Processing Devices: CPU, Memory
  - Storage Devices: Floppy Disk, Hard Disk, CD-ROM, Tape Drive, Zip Drive
  - Output Devices: Monitor, Printer, Sound Card, Modem
- **Software:** Set of instructions (programs) that control hardware operations
  - **Applications Software:** Programs for specific tasks like word processing, spreadsheets, database management, and presentations. E.g. MS Word, Excel, PowerPoint, MySQL
  - **Systems Software:** The ***operating system*** and utilities that manage hardware and enable other programs to run. E.g. Windows, Linux, macOS

# Functions of an Operating System

- **Booting the Computer:** Initializes hardware and allows the system to start
- **User Interface:** Provides Command Line or Graphical User Interface (CLI or GUI)
- **Hardware Management:** Controls and manages hardware resources
- **File System Management:** Organizes and maintains files on the disk
- **Program Support:** Provides services like saving/loading files, memory allocation, etc

# Categories of Operating Systems

- **Multitasking Operating System:**
  - Cooperative Multitasking: Requires cooperation between OS and applications (e.g., Macintosh OS)
  - Preemptive Multitasking: OS controls CPU time and priority (e.g., UNIX, Windows NT)
- **Multi-user Operating System:**
  - Supports multiple users at the same time (e.g., UNIX, Windows NT)
- **Multiprocessing Operating System:**
  - Asymmetric: One CPU controls others
  - Symmetric: No single controlling CPU, all CPUs work independently.

# Types of Operating Systems

- **DOS (Disk Operating System):** Command-line based, popular on Intel PCs
- **Microsoft Windows:** GUI-based, runs DOS and Windows programs
- **OS/2:** IBM/Microsoft OS for Intel processors, multitasking capabilities
- **Macintosh OS:** Purely graphical, easy-to-use, ideal for desktop publishing
- **UNIX:** Multi-platform, robust, command-line-based, widely used
- **Windows NT:** 32-bit OS, multitasking, supports Windows and some UNIX applications

# Principles of Networking

- **Definition of Networks**: Systems formed by links that allow information sharing. Examples include:
  - Mail delivery system
  - Telephone system
  - Corporate computer networks
  - The Internet
- **Hosts and Peripherals:**
  - A host is any device that sends/receives information on a network.
  - Devices like printers can act as peripherals or hosts depending on how they are connected.

- **Types of Devices on a Network:**
  - Desktop computers, Laptops, Tablets, Smartphones, Printers, Servers, Game consoles, Home appliances
- **Resources Shared Over Networks:**
  - Printing or scanning services
  - Storage (hard drives, optical drives)
  - Applications (e.g., databases)
  - Information (stored on other computers)
  - Documents
- **Types of Network:**
  - Personal Area Network (PAN)
  - Local Area Network (LAN)
  - Wide Area Network (WAN)
  - Metropolitan Area Network (MAN)

# Overview of the Internet

- **Definition:** A global network of computers that communicate seamlessly
- **Origins:** Began with ARPANET in the 1960s by the U.S. Department of Defense
- **Connection:** Networks worldwide communicate using Transmission Control Protocol/Internet Protocol (TCP/IP)
- **How Information is Transmitted on the Internet:**
    - Data is broken into packets for transmission.
    - TCP/IP ensures proper delivery of these packets across networks.

# Services Offered on the Internet

- **Electronic Mail (E-mail):**
  - Allows asynchronous communication (messages sent without needing the recipient to be online)
  - Cost-effective, reliable, and efficient
- **World Wide Web (WWW):**
  - Launched in 1991; uses hyperlinks for easy navigation
  - Multimedia content (text, images, video, audio)
  - Allows access to web pages through browsers

# Benefits of the Internet

- **Speed:** Quick transmission of information saves time and money.
- **Breadth:** Access to vast resources and information.
- **Cost-saving:** Cheaper than using traditional methods like fax or phone.
- **Two-way Communication:** Allows instant feedback and interaction.
- **Trading:** Supports e-commerce with:
  - Faster response times
  - Extended market reach
  - Improved customer service